

1

UNWRAPPING THE GIFT

1.1 The Pace of Change

1.2 Change and Unexpected Developments

1.3 Themes

1.4 Ethics

Exercises



1.1 The Pace of Change

In a way not seen since Gutenberg's printing press that ended the Dark Ages and ignited the Renaissance, the microchip is an epochal technology with unimaginably far-reaching economic, social, and political consequences.

—Michael Rothschild¹

In 1804, Meriwether Lewis and William Clark set out on a two-and-a-half-year voyage to explore what is now the western United States. Many more years passed before their journals were published. Later explorers did not know that Lewis and Clark had been there before them. Stephen Ambrose points out in his book about the Lewis and Clark expedition, *Undaunted Courage*, that information, people, and goods moved no faster than a horse—and this limitation had not changed in thousands of years.² In 1997, millions of people went to the World Wide Web to watch a robot cart called Sojourner roll across the surface of Mars. We chat with people thousands of miles away, and instantly view Web pages from around the world. We can tweet from airplanes flying more than 500 miles per hour.

Telephones, automobiles, airplanes, radio, household electrical appliances, and many other marvels we take for granted were invented in the late 19th and early 20th centuries. They led to profound changes in how we work and play, how we get information, how we communicate, and how we organize our family lives. Our entry into space was one of the most dramatic feats of technology in the 20th century. Sputnik, the first man-made satellite, launched in 1957. Neil Armstrong walked on the moon in 1969. We still do not have personal spacecraft, vacation trips to the moon, or a large amount of commercial or research activity in space. Space tourism for the very rich is in an early stage. The moon landing has had little direct effect on our daily lives. But computer systems in cars can now apply the brakes if a pedestrian is in the car's path. Some cars park themselves, and experimental cars drive themselves on city streets. Computer programs beat human experts at chess and *Jeopardy!*, and our smartphones answer our questions. Surgeons perform surgery with robotic instruments miles from the patient. Roughly five billion people use cellphones; U.S. texters send more than a trillion texts in a year; Facebook has more than 800 million members; Twitter users tweet hundreds of thousands of times a day; and these numbers will be out of date when you read them. A day without using an appliance or device containing a microchip is as rare as a day without turning on an electric light.

The first electronic computers were built in the 1940s. Scientists at Bell Laboratories invented the transistor—a basic component of microprocessors—in 1947. The first hard-disk drive, made by IBM in 1956, weighed more than a ton and stored only five megabytes of data, less than the amount of space we use for one photo. Now, we can walk around

with 150 hours of video in a pocket. A disk with a terabyte (one thousand gigabytes, or one trillion bytes) of storage—enough for 250 hours of high definition video—is inexpensive. There are hundreds of billions of gigabytes of information on the Internet. The 1991 space shuttle had a 1-megahertz* computer onboard. Ten years later, some luxury automobiles had 100-megahertz computers. Speeds of several gigahertz are now common. When I started my career as a computer science professor, personal computers had not yet been invented. Computers were large machines in air-conditioned rooms; we typed computer programs onto punched cards. If we wanted to do research, we went to a library, where the library catalog filled racks of trays containing 3×5 index cards. Social-networking sites were neighborhood pizza places and bars. The point is not that I am old; it is the speed and magnitude of the changes. The way you use computer systems and mobile devices, personally and professionally, will change substantially in two years, in five, and in ten, and almost unrecognizably over the course of your career. The ubiquity of computers, the rapid pace of change, and their myriad applications and impacts on daily life characterize the last few decades of the 20th century and the beginning of the 21st.

It is not just the technology that changes so fast. Social impacts and controversies morph constantly. With PCs and floppy disks came computer viruses and the beginnings of a huge challenge to the concept of copyright. With email came spam. With increased storage and speed came databases with details about our personal and financial lives. With the Web, browsers, and search engines came easy access by children to pornography, more threats to privacy, and more challenges to copyright. Online commerce brought bargains to consumers, opportunities to entrepreneurs, and identity theft and scams. Cellphones have had so many impacts that we discuss them in more detail later in this chapter and in Chapter 2. With hindsight, it might seem odd that people worried so much about antisocial, anticomunity effects of computers and the early Internet. Now, with the popularity of social networking, texting, and sharing video, photos, and information, the Net is a very social place. In 2008, “experts” worried the Internet would collapse within two years because of the demands of online video. It did not. Privacy threats of concern several years ago seem minor compared to new ones. People worried about how intimidating computers and the Internet were; now toddlers operate apps on tablets and phones. Concerns about technology “haves” and “have-nots” (the “digital divide”) waned as Internet access and cellphones spread throughout the United States and around the world, shrinking the digital divide far faster than long-standing global divides in, say, education and access to fresh water.

Discussions of social issues related to computers often focus on problems, and indeed, throughout this book we examine problems created or intensified by computer technologies. Recognizing the benefits is important too. It is necessary for forming a reasonable, balanced view of the impact and value of the technology. Analyzing and evaluating the

* This is a measure of processing speed. One megahertz is 1 million cycles per second; 1 gigahertz is 1 billion cycles per second. “Hertz” is named for the 19th-century physicist Heinrich Rudolf Hertz.

impact of new technologies can be difficult. Some of the changes are obvious. Some are more subtle. Even when benefits are obvious, the costs and side effects might not be, and vice versa. Both the technological advances brought about by computer technology and the extraordinary pace of development have dramatic, sometimes unsettling, impacts on people's lives. To some, this is frightening and disruptive. They see the changes as dehumanizing, reducing the quality of life, or as threats to the status quo and their well-being. Others see challenging and exciting opportunities. To them, the development of the technology is a thrilling and inspiring example of human progress.

When we speak of computers in this book, we include mobile devices such as smartphones and tablets, desktop computers and mainframes, embedded chips that control machines (from sewing machines to oil refineries), entertainment systems (such as video recorders and game machines), and the “Net,” or “cyberspace.” Cyberspace is built of computers (e.g., Web servers), communication devices (wired and wireless), and storage media, but its real meaning is the vast web of communications and information that includes the Internet and more.

In the next section, we look at some phenomena, often unplanned and spontaneous, that computer and communication technology made possible. They have deeply changed how we interact with other people, what we can accomplish, and how others can intrude into our relationships and activities. In the rest of the chapter, we introduce themes that show up often, and we present an introduction to some ethical theories that can help guide our thinking about controversies throughout the rest of the book. The next seven chapters look at ethical, social, and legal issues primarily from the perspective of any person who lives and works in a modern computerized society and is interested in the impact of the technology. The final chapter takes the perspective of someone who works as a computer professional who designs or programs computer systems or as a professional in any area who must make decisions and/or set policy about the use of computer systems. It explores the ethical responsibilities of the professional. The Software Engineering Code of Ethics and Professional Practice and the ACM Code of Ethics and Professional Conduct, in Appendix A, provide guidelines for professionals.

1.2 Change and Unexpected Developments

No one would design a bridge or a large building today without using computers, but the Brooklyn Bridge, built more than 130 years ago—long before computers, is both a work of art and a marvelous feat of engineering. The builders of the Statue of Liberty, the Pyramids, the Roman aqueducts, magnificent cathedrals, and countless other complex structures did not wait for computers. People communicated by letters and telephone before text messages, email, and Twitter. People socialized in person before social-networking sites. Yet we can identify several phenomena resulting from computer

and communication technology that are far different from what preceded them (in degree, if not entirely in kind), several areas where the impacts are dramatic, and many that were unanticipated. In this section, we consider a brief sampling of such phenomena. Some are quite recent. Some are routine parts of our lives now. The point is to remind us that a generation ago they did not exist. They illustrate the amazingly varied uses people find for new tools and technologies.

It is precisely this unique human capacity to transcend the present, to live one's life by purposes stretching into the future—to live not at the mercy of the world, but as a builder and designer of that world—that is the distinction between human and animal behavior, or between the human being and the machine.

—Betty Friedan³

1.2.1 CONNECTIONS: CELLPHONES, SOCIAL NETWORKING, AND MORE

The Web, social networking, cellphones, and other electronic devices keep us connected to other people and to information all day, virtually everywhere. We look at a few connectivity applications, focusing on fast changes and unanticipated uses and side effects (good and bad). The discussion suggests issues we study throughout the book.

Cellphones

In the 1990s, relatively few people had cellphones. Business people and sales people who often worked outside their office carried them. High-tech workers and gadget enthusiasts liked them. Others bought the phones so they could make emergency calls if their cars broke down. We were used to being out of touch when away from home or office. We planned ahead and arranged our activities so that we did not need a phone when one was not available. Within a short time, however, cell service improved and prices dropped. Cellphone makers and service providers developed new features and services, adding cameras, video, Web connections, and location detection. Apple introduced the iPhone in 2007, and phones got “smart.” People quickly developed hundreds of thousands of applications and embraced the term *app*. Consumers downloaded 10 billion apps from Apple’s App Store. Within very few years, people all over the world used phones, rather than PCs or laptops, as their connection to the Internet. Millions, then hundreds of millions, then billions of people started carrying mobile phones. In 2011, there were approximately five billion cellphone subscriptions worldwide—an astoundingly fast spread of a new technology. Writers describe the dramatic changes with observations such as, “A Masai warrior with a smartphone and Google has access to more information than the President did 15 years ago” and “More folks have access to a cellphone than to a toilet.”⁴

Cellphones became a common tool for conversations, messaging, taking pictures, downloading music, checking email, playing games, banking, managing investments, finding a restaurant, tracking friends, watching videos. Smartphones serve as electronic wallets and identification cards at store terminals or security checkpoints. Phones monitor security cameras at home or control home appliances from a distance. Professional people use smartphone apps for a myriad of business tasks. Smartphones with motion detectors remind obese teenagers to get moving. An app analyzes blood glucose levels for diabetics and reminds them when to exercise, take medication, or eat something. Military personnel on the front lines can use specialized apps to download satellite surveillance video and maps. More unanticipated uses include location tracking, sexting, life-saving medical apps, and malicious data-stealing apps. People use cellphones to organize flash mobs for street dances and pillow fights—or for attacking pedestrians and looting stores. Terrorists use cellphones to set off bombs. Apps designed for poor countries inform people when water is available and help perform medical imaging.

These examples suggest the number and variety of unanticipated applications of this one, relatively new “connection” device. The examples also suggest problems. We discuss privacy invasion by data theft and location tracking in Chapter 2. In Chapter 3, we consider whether phone service should be shut down during riots. Is the security of smartphones sufficient for banking and electronic wallets? (What if you lose your phone?) Do people realize that when they synch their phone with other devices, their files become vulnerable at the level of the weakest security?

As a side effect of cellphone use and the sophistication of smartphones, researchers are learning an enormous amount about our behavior. Laws protect the privacy of the content of our conversations, but smartphones log calls and messages and contain devices that detect location, motion, direction, light levels, and other phones nearby. Most owners carry their phones all day. Researchers analyze this trove of sensor data. (Yes, much of it can be stored.) Analysis of the data generates valuable information about traffic congestion, commuting patterns, and the spread of disease. In an example of the latter, by studying movement and communication patterns of MIT students, researchers could detect who had the flu, sometimes before the students knew it themselves. Researchers also can determine which people influence the decisions of others. Advertisers and politicians crave such information. Perhaps the eeriest result is that researchers who analyzed time and location data from millions of calls said that, with enough data, a mathematical model could predict where someone would be at a particular future time with more than 90% accuracy. Who will have access to that information?⁵

Rudeness is an issue with cellphones. People use them in inappropriate places, disturbing others. The fact that so many people carry small cameras everywhere (mostly in phones, but also hidden in other small objects such as pens*) affects our privacy in public

* At least one company sells a working pen that records high-resolution video.

and nonpublic places.⁶ How well do people armed with cellphone cameras distinguish news events and evidence of crimes from voyeurism, their own rudeness, and stalking?

Talking on a phone while driving a car increases the risk of an accident. Some states prohibit use of handheld phones while driving (and a lot of drivers ignore the ban). Researchers developed an app that uses motion detection by smartphones to deduce that a phone is in a moving car and block incoming calls. A more sophisticated version locates the phone well enough to block only the driver's phone, not that of a passenger.

Here is an example of a subtle behavioral change. When people began carrying cellphones and could call for help, more headed out in the wilderness or went rock climbing without appropriate preparation. In many areas of life, people take more risk when technology increases safety. This is not unreasonable if the added risk and increased safety are in balance. When rescue calls surged, some rescue services began billing for the true cost of a rescue—one way to remind people to properly weigh the risk.

Kill switches

Soon after Amazon began selling electronic books for its Kindle ebook readers, the company discovered that a publisher was selling books in Amazon's online store that it did not have the legal rights to sell in the United States. Amazon deleted the books from its store and from the Kindles of people who had bought them; it refunded their payments. A reasonable and appropriate response? Not to many customers and media observers. Customers were outraged that Amazon deleted books from their Kindles. People were startled to learn that Amazon *could* do so.* The response was so strong that Amazon announced that it would not remove books from customer Kindles again. Few realized at that time that Apple's iPhones already had a *kill switch*—a way for Apple to remotely delete apps from phones. In 2011, when a software developer discovered malicious code in an app for Android phones, Google quickly removed the app from its store and from more than 250,000 phones. Although this was a good example of the purpose of a kill switch and a beneficial use, the fact that Google could do it disturbed people. One of the troubling side effects of our connectivity is that outsiders can reach into our devices and delete our stuff.

Perhaps this extended reach should not have been a surprise. In many businesses, the IT department has access to all desktop computers and can install—or delete—software. Software on personal computers and other electronic devices communicates with businesses and organizations regularly, without our direct command, to check for updates of software, news, and our friends' activities. When we enable updates of software, a company remotely deletes old versions.

Now, the operating systems for smartphones, tablets, and some computers (e.g., Windows) have kill switches. The companies do not disclose much information about

* Ironically, one of the books Amazon removed was George Orwell's *1984*—a novel about a totalitarian government that regularly sent documents down a "memory hole" to destroy them.

them. The main purpose is security—to remove malicious software that the company discovers in an app after users have downloaded it. Indeed, companies such as Google and Apple that provide popular app stores see it as a serious responsibility to protect users from malicious apps. Some companies tell us about their removal capability in their terms of use agreements, but such agreements can run to thousands of words and have vague, general statements. Few people read them.

What are some potential uses and risks? Kill switches could remove content that infringes copyrights. They could remove content that a company or government deems offensive. What if malicious hackers found a way to operate the kill switches on our devices? Governments in many countries have extensive censorship laws and require that communications services provide government access to communications. Governments, in free and unfree countries, pressure businesses to act as the government prefers. For more than 2000 years, governments and religious and social organizations have burned books that displeased them. What pressures might governments put on companies to use the kill switches? Will the impact of electronic kill switches be more devastating than attempts to prohibit printed material? Or will companies use them carefully for improved security? Our new tools are remarkably powerful and remarkably vulnerable.

Social networking

While all this razzle-dazzle connects us electronically, it disconnects us from each other, having us “interfacing” more with computers and TV screens than looking in the face of our fellow human beings. Is this progress?

—Jim Hightower, radio commentator, 1995⁷

Facebook, one of the first of the social networking sites, started at Harvard as an online version of the hardcopy student directories available at many colleges. At first, the sites were wildly popular with young people, while older people did not understand the appeal or worried about safety and privacy. Adults quickly discovered benefits of personal and business social networking. Social networks are enormously popular with hundreds of millions of people because of the ease with which they can share so many aspects of their lives and activities with family, friends, co-workers, and the public.

As with so many other digital phenomena, people found unanticipated uses of social networking, some good, some bad. Friends and ex-boyfriends and ex-girlfriends post pranks and embarrassing material. Stalkers and bullies stalk and bully. Politicians, advertisers, businesses, and organizations seek donations, volunteers, customers, and connections. Protesters organize demonstrations and revolutions. Jurors tweet about court cases during trials (causing mistrials, overturned convictions, and jail time for offending jurors). Social networking brought us more threats to privacy and a steady stream of updates on the triv-



Privacy issues for social networks: Section 2.3.2

ial details of people's lives. Gradually, social network companies developed sophisticated privacy controls and feedback systems to reduce problems, though they certainly have not eliminated them. Overall, to most people, the benefits outweigh the problems, and social networking has become the new way of communicating.

In a phenomenon called “crowd funding,” social networks, Twitter, and other platforms make it easy to raise money in small amounts from a large number of people for charities, political causes, artistic projects, and investment in start-up companies.

How do social networking sites affect people and relationships? People can have hundreds of friends and contacts, but have they traded quality of in-person relationships for quantity of superficial digital relationships? Does the time spent online reduce the time spent on physical activity and staying healthy? It is still too early for definitive answers, but it appears that the many critics who anticipated a serious problem of social isolation were mistaken. Researchers find that people use social networks mostly to keep in touch with friends and family and that the easy, frequent contact enhances relationships, empathy, and a sense of community. On the other hand, young people who spend a lot of time on a social network do poorly in school and have behavioral problems. (Are these people who would have problems in any case? Does the access to the networks exacerbate preexisting emotional problems?)

Just as researchers study social phenomena using the masses of data that smartphone systems collect, they also mine the masses of data in social networks. For example, social scientists and computer scientists analyze billions of connections to find patterns that could help identify terrorist groups.⁸



More about artificial intelligence: Section 1.2.3

A person you follow in social media might not be a person at all. A *socialbot* is an artificial intelligence program that simulates a human being in social media. Researchers tricked Twitter users into building relationships with artificial tweeting personalities, some of which gained large followings. Political activists launched socialbots to influence voters and legislators. The U.S. military raised concerns about automated disinformation campaigns by enemies. Advertising bots are likely to be common. When the Internet was new, someone commented (and many repeated) that “on the Internet, no one knows you’re a dog.” It meant that we could develop relationships with others based on common interests without knowing or caring about age, race, nationality, gender, or physical attractiveness. Some of those others might not even be people, and we might not know it. Should we be comfortable with that?

Communication and the Web

Email and the Web are so much a part of our culture now that we might forget how new and extraordinary they are. Email was first used mostly by computer scientists. In the 1980s, messages were short and contained only text. As more people and businesses connected to computer networks, use of email expanded to science researchers, then to

businesses, then to millions of other people. Limits on length disappeared, and we began attaching digitized photos and documents. People worldwide still send several billion emails daily (not counting spam), although texting, tweeting, and other social media have replaced email as the favored communication method in many contexts.⁹

High-energy physicists established the World Wide Web in Europe in 1990 to share their work with colleagues and researchers in other countries. In the mid- and late 1990s, with the development of Web browsers and search engines, the Web became an environment for ordinary users and for electronic commerce. Today there are billions of Web pages. The Web has grown from an idea to a huge library and news source, a huge shopping mall, an entertainment center, and a multimedia, global forum in less than one generation.

The Web gives us access to information and access to audiences unimaginable a generation ago. It empowers ordinary people to make better decisions about everything from selecting a bicycle to selecting medical treatments. It empowers us to do things that we used to rely on experts to do for us. Software tools, many available for free, help us analyze the healthiness of our diet or plan a budget. We can find references and forms for legal processes. We can read frank reviews of cameras, clothing, cars, books, and other products written by other buyers, not marketing departments. We can select our entertainment and watch it when we want to. We can fight back against powerful institutions by shaming them with videos that go viral* (see, for example, “United Breaks Guitars” on YouTube) or by posting legal documents intended to intimidate us (see, for example, chillingeffects.org). Businesses and organizations use “viral marketing”—that is, relying on large numbers of people to view and spread marketing messages in clever videos. We can start our own Web-based television network without the huge investment and government license requirements of broadcast television networks. A college student with a good idea and some well-implemented software can start a business that quickly grows to be worth millions or billions of dollars; several have. The openness of the Internet enables “innovation without permission,” in the words of Vinton Cerf, one of the key people who has worked on Internet development since it began.¹⁰

Blogs (a word made up from “Web log”) and videos are two examples of the many new forms of creativity that flourish because Web technology and special software make them so easy and inexpensive. They began as outlets for amateurs and now are significant sources of news and entertainment. They have created new paths for jobs—with news media, publishers, and advertising and entertainment companies. Of course, some amateur blogs and videos are dull, silly, and poorly written or made, but many are gems, and people find them. People blog on current events, celebrity gossip, hobbies, books, movies, dieting, law, economics, technology, political candidates, Internet issues, and virtually any other topic. They provide varied, sometimes quirky perspectives. The independence of

* “Going viral” describes the phenomenon where something posted in cyberspace catches the attention of people who view, copy, and spread it (or links to it) to millions more people.

“I’ve got pressure”

When asked by a young man to speak more quietly on his cellphone, a Hong Kong bus rider berated the man for nearly six minutes with angry insults and obscenities. In the past, a few other riders might have described the incident to friends, then soon forgotten it. But in this instance, another rider captured the scene on his cellphone. The video soon appeared on the Internet, and millions of people saw it. People provided subtitles in different languages, set the video to music, used clips as mobile-phone ringtones, and

produced t-shirts with pictures and quotes. “I’ve got pressure” and other phrases from the rant slipped into conversations.

This incident reminds us that anything we do in a public place can be captured and preserved on video. But more, it illustrates how the Internet facilitates and encourages creativity and the quick creation and distribution of culture artifacts and entertainment, with the contribution of ideas, modifications, variations, improvements, and new works from thousands of people.

bloggers attracts readers; it suggests a genuine connection with what ordinary people are thinking and doing, not filtered through major news companies or governments. Businesses were quick to recognize the value of blogs, and many provide their own as part of their public relations and marketing programs. Inexpensive video cameras and video-manipulation tools have powered a burst of short amateur videos—often humorous, sometimes worthless, and sometimes quite serious. We can see a soldier’s view of war, someone’s encounter with aggressive whales, an arrest by police. Video sites also made it easy to post and trade professional videos, infringing copyrights owned by entertainment companies and individuals. We explore copyright issues in Chapter 4.

The Web connects students and teachers. At first, universities offered online courses within their area, benefitting people who work full-time, who have varying work schedules that conflict with normal class schedules, who have small children at home, or who cannot travel easily because of disabilities. Gradually a potential to revolutionize advanced education became clear.* More than 100 million people have viewed the thousands of free lessons on sciences, economics, and other subjects at the online Khan Academy. When two artificial intelligence experts offered a Stanford University graduate course for free online, they expected 500–1000 students to sign up. They got 160,000 people from around the world, and more than 20,000 completed the course, which included automatically graded homework assignments and exams.¹¹

The impact of the connections provided by the Web and cellphones is more dramatic in remote or less developed areas of the world, many of which do not have landline telephones. Mountains and thick jungle, with no roads, separate villagers in one town in

* For elementary education, it appears that regular classes and in-person teachers still have the advantage.

Telemedicine

Telemedicine, or long-distance medicine, refers to remote performance of medical exams, analyses, and procedures using specialized equipment and computer networks. On long airplane flights, telemedicine can help treat a sick passenger and ascertain whether the plane needs to make an emergency landing. Prisons use telemedicine to reduce the risk of escape by dangerous criminals. Some small-town hospitals use video systems to consult with specialists at large medical centers—eliminating the expense, time, and possible health risk of transporting the patient to the medical center. A variety of health-monitoring devices send

their readings from a patient's home to a nurse over the Internet. This technology eliminates the expense, time, and inconvenience of more frequent visits, while enabling more regular monitoring of patients and helping to catch dangerous conditions early.

Telemedicine goes well beyond transmission of information. Surgeons in New York used video, robotic devices, and high-speed communication links to remotely remove a gall bladder from a patient in France. Such systems can save lives in emergencies and bring a high level of surgical skills to small communities that have no surgeons.

Malaysia from the next, but the villagers order supplies, check the market price of rice to get a good deal when selling their crop, and email family photos to distant relatives. Farmers in Africa get weather forecasts and instruction in improved farming methods. An Inuit man operates an Internet service provider for a village in the Northwest Territories of Canada, where temperatures drop to -40°F . Villagers in Nepal sell handicrafts worldwide via a website based in Seattle. Sales have boomed, more villagers have regular work, dying local arts are reviving, and some villagers can now afford to send their children to school.

The Web abounds with examples of collaborative projects, some organized, such as Wikipedia* (the online encyclopedia written by volunteers), some spontaneous. Scientists collaborate on research with scientists in other countries much more easily and more often than they could without the Internet. Informal communities of programmers, scattered around the world, create and maintain free software. Informal, decentralized groups of people help investigate online auction fraud, a murder, stolen research, and other crimes. People who have never met collaborate on creating entertainment.

Some collaborative projects can have dangerous results. To reduce the flow of illegal immigrants, a governor of Texas proposed setting up night-vision webcams along the Mexican border that volunteers would monitor on the Internet. Will the people monitoring a border webcam go out and attack those they see coming across the border? What training or selection process is appropriate for volunteers who monitor these security cameras? In China, a man posted the online name of another man he believed was having

* A *wiki* is a website, supported by special software, that allows people to add content and edit content that others provide. Wikis are tools for collaborative projects within a business or organization or among the public.

an affair with his wife. Thousands of people participated in tracking down the man's real name and address and encouraging public action against him. Thousands of Twitterers in Saudi Arabia called for the execution of a young writer who they believed insulted the Prophet Muhammad. Mobs and individuals emotionally involved in a political, religious, or moral cause do not always pause for the details of due process. They do not carefully determine whether they identified the correct person, whether the person is guilty of a crime, and what the appropriate punishment is. On the other hand, police departments in cities in several countries effectively use instant messaging to alert residents who help find crime suspects or stolen cars in their neighborhoods. Enlisting volunteers is a useful new collaborative tool for crime fighting and possibly antiterrorism programs. How can we guide the efforts of thousands of individuals toward useful ends while protecting against mistakes, instant vigilantism, and other abuses?

1.2.2 E-COMMERCE AND FREE STUFF

In the 1990s, the idea of commercial websites horrified Web users. The Web, they believed, was for research, information, and online communities. A few brick-and-mortar businesses and a few young entrepreneurs recognized the potential and benefits of online commerce. Among the earliest traditional businesses on the Web, United Parcel Service and Federal Express let customers check the status of packages they sent. This was both a novelty and a helpful service. Amazon.com, founded in 1994, started selling books on the Web and became one of the most popular, reliable, and user-friendly commercial sites. Many, many Web-based businesses followed Amazon, creating new business models—such as eBay with its online auctions. Traditional businesses established websites. Online sales in the United States now total hundreds of billions of dollars a year. The Web changed from a mostly academic community to a world market in little more than a decade.

Some of the benefits of e-commerce are fairly obvious: we can consider more products and sellers, some far away, in less time and without burning gasoline. Some benefits are less obvious or were not obvious before they appeared. Auction sites gave people access to customers they could not have found efficiently before. The lower overhead and the ease of comparison shopping on the Web brought down prices of a variety of products. Consumers save 10–40%, for example, by buying contact lenses online, according to a Progressive Policy Institute report. Consumers who do price-comparison research on the Web before buying a new car typically save about \$400.¹² Small businesses and individual artists sell on the Web without paying big fees to middlemen and distributors. The Web enabled a peer-to-peer economy with websites where ordinary people sell or trade their skills, make small loans, and trade their homes for vacations.

Growth of commerce on the Web required solutions to several problems. One was trust. People were reluctant to give their credit card numbers on the Web to companies they had not dealt with or even heard of before. Enter PayPal, a company built on the idea of having a trusted intermediary handle payments. Encryption and secure servers also



Impacts of e-commerce
on free speech: Section
3.2.5

made payments safer.* The Better Business Bureau established a website where we can find out if consumers have complained about a company. Auction sites implemented rating and comment systems to help buyers and sellers determine whom to trust. Email confirmations of orders, consumer-friendly return policies, and easy packaging for returns all contributed to consumer comfort and more online sales. The University of Michigan's National Quality Research Center found that e-commerce businesses had a higher customer-satisfaction rating than any other sector of the economy. As online sales increased, competition led traditional stores to adopt some of the practices of e-commerce, such as consumer-friendly return policies.

Free stuff

Libraries have provided free access to books, newspapers, and journals for generations, and radio and television provided free news and entertainment before the Internet. But there is so much more free stuff now—a truly astounding amount—conveniently available on the Web.

For our computers, we can get free email programs and email accounts, browsers, filters, firewalls, encryption software, word processors, spreadsheets, software for viewing documents, software to manipulate photos and video, home inventory software, antispyware software, antivirus software, antispyware software, and software for many other specialized purposes. This is a small sampling of software available for free.

We can find free game-playing programs for old board games and card games such as chess and bridge, as well as for new games. Phone service via Skype is free. There are free dating services on the Web. Major music festivals offer their concerts for free on the Internet, a nice alternative to paying \$30 to \$500 for a ticket. Craigslist, the classified ad site, one of the most popular websites in the world, is free to people who place ads and people who read them. Major (expensive) universities such as Stanford, Yale, and MIT provide video of lectures, lecture notes, and exams for thousands of their courses on the Web for free. We can download whole books from Google, Project Gutenberg, and other sources for free.† We can read news from all over the world for free. We can store our personal photographs, videos, and other files online for free. MySpace, Facebook, Twitter, LinkedIn, and YouTube are free; Google, Bing, and Yahoo are free. Specialized, scholarly encyclopedias (e.g., the Stanford Encyclopedia of Philosophy), Wikipedia, and hundreds of other references are free.

We pay for libraries with taxes. Advertisers pay for broadcasting radio and television programs. On the Web, advertising pays for many, many free sites and services, but far from all. Wikipedia carries no advertising—donations pay for its hardware and band-

* The ease and security of payment on the Web had a pleasant side effect: Many people contribute more to charitable organizations. That had the unpleasant side effect of spawning scam charity sites.

† Books available for free downloading are in the public domain (that is, out of copyright).

width. Craigslist charges fees of some businesses that post job announcements and brokers who post apartment listings in a few cities. That keeps the site free to everyone else and free of other paid ads. Businesses provide some free information and services for good public relations and as a marketing tool. (Some free programs and services do not have all the features of the paid versions.) Nonprofit organizations provide information as a public service; donations or grants fund them. One of the distinct and delightful features of the Internet is that individuals provide a huge amount of free stuff simply because it pleases them to do so. They are professionals or hobbyists or just ordinary people who enjoy sharing their expertise and enthusiasm. Generosity and public service flourish in the Web environment.

It is often obvious when we are viewing advertisements on websites or phones. Ads annoy some people, but they are not insidious, and their presence on a screen is not an unreasonable price to pay for free services. However, to earn ad revenue to fund multimillion-dollar services, many free sites collect information about our online activities and sell it to advertisers. This tracking is often not obvious; we consider it in Chapter 2.

1.2.3 ARTIFICIAL INTELLIGENCE, ROBOTICS, SENSORS, AND MOTION

Artificial intelligence

Artificial intelligence (AI) is a branch of computer science that makes computers perform tasks we normally (or used to) think of as requiring human intelligence. It includes playing complex strategy games such as chess, language translation, making decisions based on large amounts of data (such as approving loan applications), and understanding speech (where the appropriateness of the response might be the measure of “understanding”). AI also includes tasks performed automatically by the human brain and nervous system—for example, vision (the capture and interpretation of images by cameras and software). Learning is a characteristic of many AI programs. That is, the output of the program improves over time as it “learns” by evaluating results of its decisions on the inputs it encounters. Many AI applications involve *pattern recognition*, that is, recognizing similarities among different things. Applications include reading handwriting (for automatic sorting of mail and input on tablet computers, for example), matching fingerprints, and matching faces in photos.

Early in the development of AI, researchers thought the hard problems for computers were tasks that required high intelligence and advanced training for humans, such as winning at chess and doing mathematical proofs. In 1997, IBM’s chess computer, Deep Blue, beat World Champion Garry Kasparov in a tournament. AI researchers realized that narrow, specialized skills were easier for computers than what a five-year-old does: recognize people, carry on a conversation, respond intelligently to the environment. In 2011, another specially designed computer system called Watson (also built by IBM) defeated human *Jeopardy!* champions by answering questions more quickly than the humans. Watson processes language (including puns, analogies, and so on) and general

knowledge. It searches and analyzes 200 million pages of information in less than three seconds. Practical applications of the Watson technology include medical diagnosis and various business decision-making applications.

We briefly describe a few more examples of AI applications. They were astonishing advances not long ago.

When a man had a heart attack in a swimming pool in Germany, lifeguards did not see him sink to the bottom of the pool. An underwater surveillance system, using cameras and sophisticated software, detected him and alerted the lifeguards who rescued him. The software distinguishes a swimmer in distress from normal swimming, shadows, and reflections. It is now installed in many large pools in Europe and the United States. Just as AI software can distinguish a swimmer in trouble from other swimmers, AI software in video surveillance systems can distinguish suspicious behavior by a customer in a store that might indicate shoplifting or other crimes. Thus, without constant human monitoring, the AI-equipped video system can help prevent a crime, rather than simply identify the culprits afterwards.

Search engines use AI techniques to select search results. They figure out what the user meant if the search phrase contains typos, and they use context to determine the intended meaning of words that have multiple meanings. Automated websites that answer questions use AI to figure out what a question means and find answers.

Speech recognition, once a difficult research area, is now a common tool for hundreds of applications. Computer programs that teach foreign languages give instruction in correct pronunciation if they do not recognize what the user says. Millions of people who carry Apple smartphones can ask questions of Siri, Apple's "intelligent" personal assistant. Siri interprets our questions and searches the Web for answers. Air traffic controllers train in a mockup tower whose "windows" are computer screens. The trainee directs simulated air traffic. The computer system responds when the trainee speaks to the simulated pilots. Such simulation allows more intensive training in a safe environment. If the trainee mistakenly directs two airplanes to land on the same runway at the same time, no one gets hurt.

People continue to debate the philosophical nature and social implications of artificial intelligence. What does it mean for a computer system to be intelligent? Alan Turing, who developed fundamental concepts underlying computer science before there were computers, proposed a test, now called the Turing Test, for human-level intelligence. Let a person converse (over a network) with the system on any topics the person chooses. If the computer convinces the person that it is human, the computer passes the test. Is that enough? Many technologists think so (assuming the actual test is well designed). But is the computer intelligent? Philosopher John Searle argues that computers are not and cannot be intelligent. They do not think; they manipulate symbols. They do so at very high speed, and they can store (or access) and manipulate a huge quantity of data, but they are not conscious. They do not understand; they simulate understanding. Searle uses the following example to illustrate the difference: Suppose you do not know the

Chinese language. You are in a room with lots of boxes of Chinese symbols and a large instruction book written in English. People submit to you sequences of Chinese symbols. The instructions tell you how to manipulate the symbols you are given and the ones in the boxes to produce a new sequence of symbols to give back. You are very careful, and you do not get bored; you follow the instructions in the book exactly. Unknown to you, the sequences you receive are questions in Chinese. The sequences that you give back by following the instructions (just as a computer follows the instructions of a program) are the correct answers in Chinese. Everyone outside the room thinks you understand Chinese very well. Do you? Searle might say that although Watson won at *Jeopardy!*, Watson does not know it won.¹³

Whether we characterize machines as intelligent, or use the word metaphorically, or say that machines simulate intelligence, advances in AI are continuing at a very fast pace. It took IBM several years and millions of dollars to build Watson.¹⁴ Technologist Ray Kurzweil thinks personal computers will have the power of Watson within 10 years.

The goal of 17th- and 18th-century calculators was modest: to automate basic arithmetic operations. It shocked people at the time. That a mindless machine could perform tasks associated with human intellectual abilities was disconcerting. Centuries later, Garry Kasparov's loss to a computer chess program generated worried articles about the value—or loss of value—of human intelligence. Watson generated more. So far, it seems that each new AI breakthrough is met with concern and fear at first. A few years later, we take it for granted. How will we react when *Jeopardy!* is oh, so trivial that anyone can do well at it? How will we react when we can go into a hospital for surgery performed entirely by a machine? Will it be scarier than riding in the first automatic elevators or airplanes? How will we react when we can have a conversation over the Net about any topic at all—and not know if we are conversing with a human or a machine? How will we react when chips implanted in our brains enhance our memory with gigabytes of data and a search engine? Will we still be human?



Implications of human-level AI: Section 7.4.3

Robots

Robots are mechanical devices that perform physical tasks traditionally done by humans or tasks that we think of as human-like activities. Robotic machines have been assembling products in factories for decades. They work faster and more accurately than people can. Computer software with artificial intelligence controls most robotic devices now. Robotic milking machines milk hundreds of thousands of cows at dairy farms while the farmhands sleep or do other chores. Some robots dance, and some make facial expressions to convey emotions. However, just as general intelligence is a hard problem for AI, general movement and functioning is a hard problem for robots. Most robotic devices are special-purpose devices with a relatively limited set of operations.

McDonald's and other fast-food sellers use robotic food preparation systems to reduce costs and speed service. A robot pharmacist machine, connected to a patient database,

plucks the appropriate medications from pharmacy shelves by reading bar codes, checks for drug interactions, and handles billing. One of its main goals is reduction of human error. Robots deliver medications and carry linens in hospitals. They navigate around obstacles and “push” elevator buttons with wireless signals. Physicians do complex and delicate surgery from a console with a 3-D monitor and joysticks that control robotic instruments. The software filters out a physician’s shaky movements. Robots work in environments that are hazardous to people. They inspect undersea structures and communication cables. They search for survivors in buildings collapsed by bombs or earthquakes. They explore volcanoes and other planets. They move or process nuclear and other hazardous wastes.

For several years, Sony sold a robot pet dog, Aibo. It walked (with a camera system providing vision). It responded to commands, and it learned. Several companies make robots with a more-or-less human shape. Honda’s Asimo, for example, walks up and down stairs. Various companies and researchers are developing robots with more general abilities. One goal is to develop robots that can act intelligently and perform a variety of operations to assist people. Robots (doglike or humanlike) can serve as companions to elderly people. Is an emotional connection with a machine dehumanizing, or is it an improvement over living alone or in a nursing home where the staff cannot provide regular companionship? Will knowing that Grandma has a robot companion ease the guilt of family members and lead them to visit less often? Will we come to view robot companions as positively as pets?

Smart sensors, motion, and control

How do robots walk, climb stairs, and dance? Tiny motion-sensing and gravity-sensing devices collect status data. Complex software interprets the data and determines the necessary motions, and then sends signals to motors. These devices—accelerometers, or *mems* (for microelectromechanical systems)—help robots, and Segway’s motorized scooters, stay upright.

A sharp price drop for mems triggered a burst of applications.¹⁵ They provide image stabilization in digital cameras. They detect when a car has crashed, when someone has dropped a laptop, or when an elderly person has fallen. (In those applications, the system deploys an airbag, triggers a lock on the disk drive to reduce damage, or calls for help.) The Wii game console, whose controller detects the user’s motion, and motion detectors in smartphones brought motion-sensing applications to millions of consumers.

Tiny microprocessors with sensors and radio transmitters (sometimes called smart dust, though they are still larger than dust particles) are finding all sorts of applications. Some are in use; some are in development. We mention a few examples. These examples have many obvious benefits. What are some potential problems?

Oil refineries and fuel storage systems use thousands of sensors to detect leaks and other malfunctions. Sandia National Laboratory developed a “chemical lab on a chip” that can detect emissions from automobiles, chemical leaks, dangerous gases in fires (reducing

risk for firefighters), and many other hazards. Similar chips could detect chemical warfare agents.

Sensors detect temperature, acceleration, and stress in materials (such as airplane parts). Sensors distributed throughout buildings and bridges can detect structural problems, report on damage from earthquakes, and so on. These applications increase safety while reducing maintenance costs.

Sensors in agricultural fields report on moisture, acidity, and so on, helping farmers to avoid waste and to use no more fertilizer than needed. Sensors could detect molds or insects that might destroy crops. Sensors implanted in chickens monitor the birds' body temperature. A computer automatically reduces the temperature in the chicken coop if the birds get too hot, thus reducing disease and death from overheating. Sensors in food products monitor temperature, humidity, and other factors to detect potential health problems while the food is in transit to stores.

What will be the impact of tiny flying sensor/computers that communicate wirelessly and which the military can deploy to monitor movement of equipment and people, or with which police or criminals can spy on us in our homes and public places?

A Microsoft researcher developed a system with which a user manipulates 3-D images with hand movements, without touching a screen or any controls. Designers of buildings, machines, clothing, and so on, could use it to examine designs before implementing them. Someone with dirty (or sterile) hands (e.g., mechanics, cooks, surgeons) could examine reference materials while working. What other applications will people think of?

Sensors in baby clothes detect when a baby is sleeping face down, at risk for Sudden Infant Death Syndrome, and warn parents on their cellphone. A heart monitor in a firefighter's shirt alerts supervisors if the firefighter is too stressed and needs a break. Trainers plan to use sensors in special clothing to better train athletes. What other applications will we find for *wearware*?

Already we implant or attach microprocessor-controlled devices in or on human bodies: heart pacemakers and defibrillators and devices that restore motion to paralyzed people (which we describe in Section 1.2.4). These will likely see modifications that enhance performance for healthy people. At first it might be physical performance for athletes—for example, to help a competitive swimmer swim more smoothly. Then what? Biological sciences and computer sciences will combine in new ways.

1.2.4 TOOLS FOR DISABLED PEOPLE

One of the most heartwarming applications of computer technology is the restoration of abilities, productivity, and independence to people with physical disabilities.

Some computer-based devices assist disabled people in using ordinary computer applications that other people use, such as Web browsers and word processors. Some enable disabled people to control household and workplace appliances that most of us operate by hand. Some improve mobility. Some technologies that are primarily conveniences

for most of us provide significantly more benefit for disabled people: consider that text messaging was very popular among deaf people before it was popular with the general population.

For people who are blind, computers equipped with speech synthesizers read aloud what a sighted person sees on the screen. They read information embedded in Web pages that sighted visitors do not need, for example, descriptions of images. Google offers search tools that rank websites based on how accessible they are for blind users. For materials not in electronic form, a scanner or camera, optical-character-recognition software, and a speech synthesizer combine to read aloud to a blind person. The first such readers were large machines. Now, handheld versions can read menus, bills, and receipts in restaurants, as well as magazines and mail at home. Where noise is a problem (or for a person both blind and deaf), a grid of buttons raised and lowered by the computer to form Braille characters can replace speech output. Braille printers provide hard copy. (Books have long been available in Braille or on tape, but the expense of production for a small market kept the selection limited.) Systems similar to navigation systems in cars help blind people walk around and find their way in unfamiliar neighborhoods.

Prosthetic devices, such as artificial arms and legs, have improved from heavy, “dumb” wood, to lighter materials with analog motors, and now to highly sensitive and flexible digitally controlled devices that enable amputees to participate in sports and fly airplanes. A person whose leg was amputated above the knee can walk, sit, and climb stairs with an artificial “smart” knee. Sensors attached to the natural leg measure pressure and motion more than a thousand times a second and transmit the data to a processor in the prosthetic leg. Artificial intelligence software recognizes and adapts to changes in speed and slope and the person’s walking style. The processor controls motors to bend and straighten the knee and support the body’s movement, replacing the normal complex interplay of nerves, muscles, tendons, and ligaments. Artificial arms use electrodes to pick up tiny electrical fields generated by contractions of muscles in the upper (natural) limb. Microprocessors control tiny motors that move the artificial limb, open and close fingers, and so on. For people with paralyzed legs or for others who cannot use an artificial leg, there are wheelchairs that climb stairs and support and transport a person in an upright position. In 2012, Exo Bionics sold its first exoskeleton, a device with sensors and tiny motors that straps to a person with paralyzed legs and enables the person to walk.¹⁶

Various conditions—loss of limbs, quadriplegia (paralysis in both arms and legs, often resulting from an accident), and certain diseases—eliminate all or almost all use of the hands. Speech recognition systems are an extremely valuable tool for these people and for others. (Deaf people can use speech-recognition systems to “hear” another speaker as the computer displays the spoken words on a screen.) People who cannot use their hands can dictate documents to a word processor and give commands to a computer to control household appliances.

To restore control and motion to people paralyzed by spinal injuries, researchers are experimenting with chips that convert brain signals to controls for leg and arm muscles.

Researchers in the United States and Europe are developing brain–computer interfaces so that severely handicapped people can operate a computer and control appliances with their thoughts.¹⁷

The impact of all these devices on the morale of the user is immense. Think about a person with an active mind, personality, and sense of humor—but who cannot write, type, or speak. Imagine the difference when the person gains the ability to communicate—with family and friends, and with all the people and resources available on the Internet.

1.3 Themes

Several themes and approaches to analysis of issues appear through this book. I introduce a few here.

Old problems in a new context

Cyberspace has many of the problems, annoyances, and controversies of noncyber life, among them crime, pornography, violent fiction and games, advertising, copyright infringement, gambling, and products that do not work right.

Throughout this book, I often draw analogies from other technologies and other aspects of life. Sometimes we can find a helpful perspective for analysis and even ideas for solutions to new problems by looking at older technologies and established legal and social principles. The emphasis on the fact that similar problems occur in other areas is not meant to excuse the new problems. It suggests, however, that the root is not always the new technology but can be human nature, ethics, politics, or other factors. We will often try to analyze how the technology changes the context and the impact of old problems.

Adapting to new technology

Changes in technology usually require adaptive changes in laws, social institutions, business policies, and personal skills, attitudes, and behavior.

When cellphones first came with built-in cameras, privacy laws in Pennsylvania (and elsewhere) were not sufficient to convict a man who used his cellphone to take a photo up a woman's skirt. (The man was found guilty of disorderly conduct.) A federal regulation requiring medical x-rays on film, rather than digital formats, was still in effect in 2011. During Japanese election campaigns in 2005, candidates were afraid to use email and blogs and to update their websites to communicate with voters, because a 1955 law that specifies the legal means of communicating with voters does not, of course, include these methods. It allows postcards and pamphlets.

We might naturally think some actions are criminal, and some should be legal, but legislators did not consider them when writing existing laws. The legal status of an action might be the opposite of what we expect, or it might be uncertain. Many new activities

that new technology makes possible are so different from prior ways of doing things that we need a new set of “rules of the game.”

We have to relearn standards for deciding when to trust what we read. The major impact of computer technology on privacy means we have to think in new ways about how to protect ourselves. We have to decide when privacy is important and when we are willing to put it at risk for some other benefit.

Varied sources of solutions to problems

Solutions for problems that result from new technology come from more or improved technology, the market, management policies, education and public awareness, volunteer efforts, and law.

The cycle of problems and solutions, more problems and more solutions, is a natural part of change and of life in general. Throughout this book, when we consider problems, we consider solutions from several categories. Technical solutions include hardware and software. “Hardware” might mean something other than part of a computer system; improved lighting near ATMs to reduce robberies is a hardware solution. Authentication technology helps reduce identity theft. Market mechanisms, such as competition and consumer demand, generate many improvements. We all must become educated about the risks of the high-tech tools we use and learn how to use them safely. Legal solutions include effective law enforcement, criminal penalties, lawsuits, legislation, and regulation. For example, there must be appropriate penalties for people who commit fraud online, and there must be appropriate liability laws for cases where system failures occur.

The global reach of the Net

The ease of communication with distant countries has profound social, economic, and political effects—some beneficial, some not.

The Net makes information and opportunities more easily available to people isolated by geography or by political system. It makes crime fighting and law enforcement more difficult, because criminals can steal and disrupt services from outside the victim’s country. Laws in one country prohibiting certain content on the Web or certain kinds of Web services restrict people and businesses in other countries because the Web is accessible worldwide.

Trade-offs and controversy

Increasing privacy and security often means reducing convenience. Protecting privacy makes law enforcement more difficult. Unpleasant, offensive, or inaccurate information accompanies our access to the Web’s vast amounts of useful information.

Some of the topics we discuss are not particularly controversial. We will sometimes address an issue more as a problem-solving exercise than as a controversy. We will look at the

impact of electronic technology in a particular area, observe some problems that result, and describe solutions. On the other hand, many of the issues are controversial: leaking confidential information on the Internet, proper policies for privacy protection, how strict copyright law should be, offshoring of jobs, the impact of computers on quality of life.

We consider various viewpoints and arguments. Even if you have a strong position on one side of a controversy, it is important to know the arguments on the other side, for several reasons. Knowing that there are reasonable arguments for a different point of view, even if you do not think they are strong enough to win overall, helps make a debate more civilized. We see that the people on the other side are not necessarily evil, stupid, or ignorant; they may just put more weight on different factors. To convince others of your own viewpoint, you must counter the strongest arguments of the other side, so, of course, you first must know and understand them. Finally, you might change your own mind after considering arguments you had not thought of before.

Perfection is a direction, not an option.

In general, when evaluating new technologies and applications, we should not compare them to some ideal of perfect service or zero side effects and zero risk. That is impossible to achieve in most aspects of life. Instead, we should compare them to the alternatives and weigh the problems against the benefits. The ideal shows us the direction to go as we endeavor to seek improvements and solutions to problems.

Another reason that we cannot expect perfection is that we all have different ideas of what perfection is.

This does not excuse sloppiness. It is possible to meet extremely high standards.

Differences between personal choices, business policies, and law

The criteria for making personal choices, for making policies for businesses and organizations, and for writing laws are fundamentally different.

We can make a personal choice—for example, about what social networks to join, what apps to put on our phones, or what ebooks to buy—according to our individual values and situation. A business bases its policies on many factors, including the manager's perception of consumer preferences, what competitors are doing, responsibilities to stockholders, the ethics of the business owners or managers, and relevant laws.

Laws are fundamentally different from personal choices and organizational policies because they impose decisions by force on people who did not make them. Arguments for passing a law should be qualitatively different from reasons for adopting a personal or organizational policy. It might seem odd at first, but arguments on the merits of the proposal—for example, that it is a good idea, or is efficient, or is good for business, or is helpful to consumers—are not good arguments for a law. We can use these arguments to try to convince a person or organization to adopt a particular policy voluntarily. Arguments for a law must show why the decision should be enforced against someone

who *does not agree* that it is a good idea. It is better to base laws on the notion of rights rather than on personal views about their benefits or how we want people to behave.

1.4 Ethics

Honesty is the best policy.

—English proverb, pre-1600

1.4.1 WHAT IS ETHICS, ANYWAY?

Sometimes, we discuss issues and problems related to computer technology from a somewhat detached perspective. We see how a new technology can create new risks and how social and legal institutions continually adapt. But technology is not an immutable force, outside of human control. People make decisions about what technologies and products to develop and how to use them. People make decisions about when a product is safe to release. People make decisions about access to and use of personal information. People make laws and set rules and standards.

Should you download movies from unauthorized websites? Should you talk on your cellphone while driving on a freeway? Should you hire foreign programmers who work at low salaries? Should you warn potential customers that the smartphone app you sell needs to copy their contact list? Should you fire an employee who is criticizing your business in social media? What information should you allow advertisers and other trackers to collect from visitors to the website you run? Someone sent you the contents of a friend's (a teacher's, a city council candidate's) email account; should you post it on the Web? In these examples, you are confronting practical and legal issues—and ethical ones. In each case you can restate the problem as a question in the form “Is it right to . . . ?” Is it right to make a significant change in your company's privacy policy without giving customers or members advance notice?

In this section, we introduce several ethical theories. We discuss some distinctions (e.g., between ethics and law) that are important to understand when tackling ethical issues.

Ethics is the study of what it means to “do the right thing.” It is a complex subject that has occupied philosophers for thousands of years. This presentation is necessarily simplified.

Ethical theory assumes that people are rational and make free choices. Neither of these conditions is always and absolutely true. People act emotionally, and they make mistakes. A person is not making a free choice when someone else is pointing a gun at him. Some argue that a person is not making a free choice in a situation where she might lose a job. However, free choice and use of rational judgment are capacities and characteristics of